

Prestige vs. Price: Spatial Capitalization of Higher Education at State Borders

Evidence from a Difference-in-Discontinuities Design (2012–2024)

Chen Chun, Jiang Xinyan, Xie Jingni

Faculty of Business and Economics, The University of Hong Kong

May 3, 2026

Roadmap

- **Setup & Hypotheses:** Spatial boundaries of public university tuition and caliber at state borders.
- **Literature & Story:** Why prestige can be priced separately from tuition savings.
- **Empirical Strategy:** The Spatial Difference-in-Discontinuities (DiDC) method.
- **Data & Curation:** Ranking variation, NBER TAXSIM microsimulations, and MSE-optimal bandwidth selection.
- **Results:**
 - ▶ Baseline
 - ▶ Heterogeneity
 - ▶ Robustness
- **Future steps**
- **Conclusion**

Background: Higher Education at the State Border

- State borders create sharp, discontinuous "cliffs" in university access.
- Crossing a state line changes one's access to:
 - ① **Financial Subsidies:** Tens of thousands of dollars in in-state tuition discounts.
 - ② **Prestige:** Potential access to elite, nationally-renowned public flagship institutions. In-state applicants generally have higher acceptance rates.
- **The Question:** Do households spatially sort to capture these benefits, and is this access capitalized into local housing markets?

Hypotheses

Hypothesis 1: Financial Cost Driven

Housing markets price the pure financial savings of the college "tuition cliff." Families sort to capture cheaper in-state pricing.

Hypothesis 2: Prestige Option Value

Housing markets price the *option value* of access to elite public flagship tiers (e.g., Top 50 institutions), regardless of the exact financial cost.

Theoretical Framework: Spatial Supply and Demand

- **Setup:** Two states (A and B) separated by a border, a local market is formed at the border.
- **Mechanism:** Housing supply $S(P)$ is fixed in the short run at exactly the border line.
- Residential demand $D(P, A_s)$ depends on housing price P and state-specific amenities $A_s = f(\text{Tuition}_s, \text{Caliber}_s)$.
- **Intuition:** If State A randomly discounts its in-state tuition or gains a Top 50 university, its amenity value A_A spikes.
- Demand shifts across the border from State B into State A. Since supply at the border is inelastic, land prices in A must rise discontinuously relative to B to clear the market.

Literature & Contribution

- **Tiebout Sorting (1956) & School Capitalization:** Black (1999) proves housing markets capitalize local K-12 school quality at attendance boundaries.
- **Value of Education:** A large economics literature studies whether households pay for human capital, peer quality, signaling value, school quality, or access to selective institutions.
- **Higher-Education Prestige:** A central education question is why households and students value selective universities so strongly.
- **The Gap:** We know little about whether families sort at the *state level* for higher education, or whether prestige itself is priced separately from tuition savings.
- **Contribution:**
 - ▶ Extends Tiebout sorting from K-12 to the higher-education market.
 - ▶ Provides revealed-preference evidence on how responsive households are to elite university prestige.
 - ▶ Overcomes policy endogeneity typical of macro-state comparisons by using a high-resolution Spatial Difference-in-Discontinuities (DiDC) estimator across 448k estimation observations.

Story: Prestige as an Option Value

- The project is not only about whether college is cheaper across the border.
- It asks whether households pay today for the future option of in-state access to elite public universities.
- That makes the housing market a revealed-preference setting for a broader education-literature question:
 - ▶ How much do households value university prestige itself?
 - ▶ Is that value visible even before a child applies to college?
- **Prediction:** If prestige is the relevant amenity, the effect should be strongest around salient elite tiers and in places where education demand is high.

Empirical Strategy: Spatial DiDC

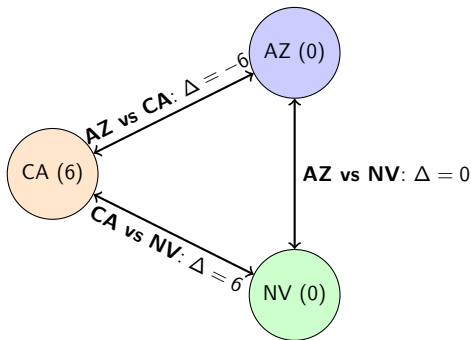
Econometric Specification

$$\ln(P_{zt}) = \beta(S_i \times \Delta\text{Edu}_{st}) + \gamma X_{zt} + \alpha_{sy} + \epsilon_{zt}$$

- S_i : Spatial dummy (1 if Treated Side, 0 if Control Side).
- ΔEdu_{st} : The dyadic gap in Tuition or Caliber.
- γX_{zt} : Time-varying controls (Federal & state income tax, property taxes, local GDP, unemployment).
- α_{sy} : **Border-Segment-by-Year Fixed Effects**. Absorbs all cross-border temporal and unobserved spatial shocks specific to that exact junction.
- **Two-Way Clustered Standard Errors** (ZCTA and Border Dyad).

Spatial Dummy Assignment

The Assignment Problem: For any state dyad, which state is $S_i = 1$? We arbitrarily sort alphabetically. The parameter math flawlessly absorbs the polarity.



- If CA is control ($S_{AZ} = 1$), $\Delta = -6$. If CA is treated ($S_{CA} = 1$), $\Delta = +6$.
- Switching the dummy assignment does not affect the β coefficient.

Data Sources & Variable Construction

Table: Data Sources and Variable Construction

Variable	Source	Construction
Housing prices	Redfin	Median price per sq. ft. (ZCTA, annual)
Tuition	IPEDS	In-state tuition, enrollment-weighted
University prestige	U.S. News	Top 50/100 public university counts by state
Income taxes	NBER TAXSIM	Simulated federal & state liability
Property taxes	ACS 5-year	Effective rate (taxes paid / home value)
Macro controls	BEA, BLS	County real GDP per capita, unemployment

Data Curation: Housing & Education

1. Housing Outcomes (2012–2024)

- ZCTA-level panel from Redfin (Median Price per Square Foot).
- Derived distances to the boundary (d_{ic}) for precise spatial sorting.

2. Higher Education Policy

- **Financial:** IPEDS panel for in-state tuition weighted by enrollment.
- **Prestige:** U.S. News historical rankings, binned into Top 20, Top 50, and Top 100 state flagship caliber indicators.

What Variation Identifies Prestige?

- The treatment is the **cross-border gap** in public university caliber.
- Baseline measure: number of public universities in a state-year ranked in the **Top 50**.
- Identification comes from **changes over time** in this gap within the same border segment.
- This is why the design does not rely only on permanent differences like California vs. Nevada.
- **Important next check:** show the underlying ranking movements directly and test alternative measures:
 - ▶ average public-university rank,
 - ▶ enrollment-weighted rank,
 - ▶ admissions-access or program-specific quality measures.

- **The Problem:** High-amenity states charge high taxes. Failing to control for tax shifts across the border thoroughly poisons the tuition coefficient.
- **The Solution (NBER TAXSIM):**
 - ▶ We simulated a standardized "Median Household" (\$100k income, married, 2 dependents).
 - ▶ Fed this identical family through the tax code of all 50 states natively for every year.
 - ▶ Isolated pure statutory tax liability variation without endogeneity from actual local wage or employment differences.

Methodology: Border Segmentation & Bandwidth

- **The Problem:** Comparing across entire state borders confounds climate and economy.
- **Segmentation:** We fracture the US state-border network into discrete, evenly distributed segments ($\approx 50\text{km}$ bands).
- **Bandwidth (h^*):** We compute data-driven, MSE-optimal geographic target bandwidths locally using Triangular Kernels (via `rdrobust`).
 - ▶ **Efficiency:** Filters 2.7M raw observations down to a 448k estimation sample (16% retention).
 - ▶ **Mean Bandwidth:** $h^* \approx 63.33$ km.
 - ▶ **Failsafe:** Global fallback of 50km applied to sparse rural segments to ensure spatial completeness.
- This forces comparisons strictly between immediate cross-state neighbors operating in identical macroeconomic geographies.

Summary Statistics (Estimation DiDC Panel)

Table: ZCTA-Level Estimation Sample Properties (Filtered to Bandwidth)

Variable	Observations	Mean	Std. Dev.
Median Price Per SqFt (\$)	268,224	163.22	367.17
Distance to Border d_{ic} (km)	448,941	0.43	36.82
Δ In-State Tuition Gap (\$)	448,941	-163.74	3390.95
Δ Top 50 Caliber Gap (Count)	344,411	-0.11	0.88
Effective Property Tax Rate	407,324	0.012	0.007
Unemployment Rate (%)	444,795	5.071	2.174
Real GDP Per Capita (\$)	444,795	58,720	1,050,974
TAXSIM Federal Income Tax (\$)	419,176	198.88	6754.56
TAXSIM State Income Tax (\$)	419,176	1959.85	1924.20

- Taxes simulated for a \$100k median household; GDP and Unemployment captured at ZCTA-level.

Result I: The "Tuition Illusion"

Table: Capitalization of Higher Education Tuition Subsidies (2012–2024)

	(1) Spatial RD	(2) Full DiDC
Tuition Gap LATE (\$1k)	−0.039* (0.022)	−0.006 (0.014)
Spatial Treatment (S_i)	−0.114* (0.068)	−0.061 (0.052)
Effective Property Tax Rate		−19.040*** (3.318)
Log Real GDP Per Capita		0.035*** (0.009)
Local Unemployment Rate		−0.048*** (0.009)
TAXSIM Covariates		✓
Fixed Effects	None	Seg × Year
Observations	184,304	118,179
R^2	0.038	0.811

- **Finding:** Raw Spatial RD assumes cheaper tuition drives higher prices (Col 1). Saturated DiDC models prove this is an illusion driven entirely by omitted state tax structures (Col 2).

Result II: Prestige Capitalization

Table: Capitalization of Elite Higher Education Institutional Caliber (Top 50, 2012–2024)

	(1) Spatial RD	(2) Full DiDC
Caliber Gap LATE (Top 50)	0.010 (0.042)	0.086** (0.036)
Spatial Treatment (S_i)	−0.128 (0.088)	−0.037 (0.054)
Effective Property Tax Rate		−19.521*** (3.293)
Log Real GDP Per Capita		0.040*** (0.010)
Local Unemployment Rate		−0.042*** (0.009)
TAXSIM Covariates		✓
Fixed Effects	None	Seg × Year
Observations	184,304	118,179
R^2	0.004	0.811

Result III: The Joint Capitalization Model

Table: Joint Capitalization Model: Tuition and Top 50 Caliber (2012–2024)

	(1) Spatial RD	(2) Full DiDC
Caliber Gap LATE (Top 50)	0.032 (0.046)	0.090*** (0.031)
Tuition Gap LATE (\$1k)	−0.040* (0.022)	−0.009 (0.014)
Effective Property Tax Rate		−19.195*** (3.273)
Log Real GDP Per Capita		0.037*** (0.010)
Local Unemployment Rate		−0.044*** (0.008)
TAXSIM Covariates		✓
Fixed Effects	None	Seg × Year
Observations	184,304	118,179
R^2	0.039	0.811

- **Concrete Magnitude:** Gaining in-state access to a Top 50 university increases local housing prices by a precisely estimated **9.0%** ($p < 0.01$) exactly at the state border.

Exploring Option-Value Non-Linearities

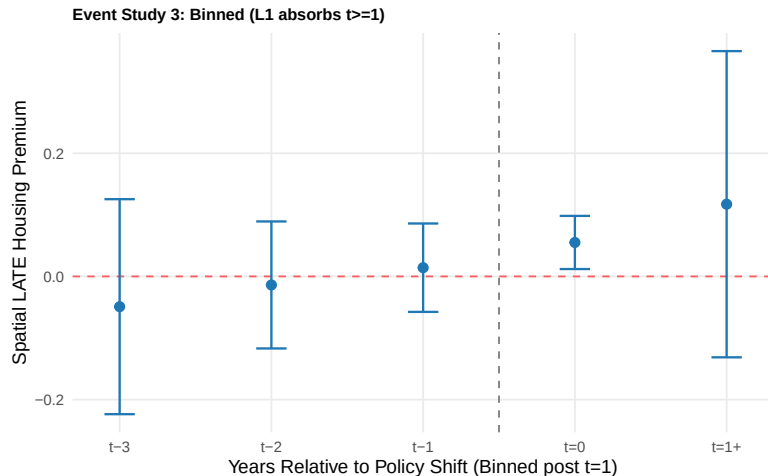
- Does the prestige tier matter? Yes. We observe a structural inverse U-shape in the willingness to pay for elite access.

Table: Option Value Non-Linearities: Caliber Tier Comparison (Two-Way Clustered SE)

Caliber Tier	Coefficient	Std. Error	Observations
Top 20 Gap	0.062	(0.047)	118,179
Top 30 Gap	0.196***	(0.062)	118,179
Top 40 Gap	0.086**	(0.034)	118,179
Top 50 Gap	0.090***	(0.031)	118,179
Top 60 Gap	0.079**	(0.031)	118,179
Top 70 Gap	0.065	(0.041)	118,179
Top 80 Gap	0.035	(0.026)	118,179
Top 100 Gap	0.026	(0.027)	118,179
Top 150 Gap	−0.000	(0.028)	118,179

- Intuition:** Top 100 is too broad to induce cross-state sorting. Top 20 is mathematically too sparse geographically to generate signal. Top 50 bounds the precise threshold of the market's psychological valuation.

Event Study: Validation & Market Efficiency



Key Takeaways:

- **Zero Pre-Trends:** Leads ($t < 0$) firmly cross zero. Prevents reverse-causality.
- **Efficient Market:** Capitalization forces resolve definitively at strictly $t = 0$, fully reflecting option-value instantly upon US News release.

Identification Threats We Need to Address

- **Main threat:** omitted variables may jointly move rankings and real estate prices.
- Example: a state economic boom could raise university resources and housing demand at the same time.
- Current responses:
 - ▶ border-segment-by-year fixed effects compare immediate cross-border neighbors,
 - ▶ TAXSIM and property-tax controls absorb large fiscal confounders,
 - ▶ event-study leads show no pre-trend before prestige gaps open.
- **Interpretation:** strong suggestive evidence of prestige capitalization, with exogeneity of ranking variation as the key remaining publication hurdle.

Caveats & Limitations

- **Local Average Treatment Effect (LATE):** The 9.0% premium is intensely localized to the immediate geospatial state border. Housing markets deep in the interior of a state possess considerably lower spatial elasticity due to commuting limits.
- **Prestige Definition:** U.S. News Top 50 is a known, highly publicized public proxy. The genuine structural option value might operate on specific major availability (e.g., highly restricted elite Engineering or Pre-Med programs) embedded solely within those Top 50 schools.
- **Ranking Exogeneity:** We still need to show more clearly why ranking movements are not caused by the same state trends that move housing prices.

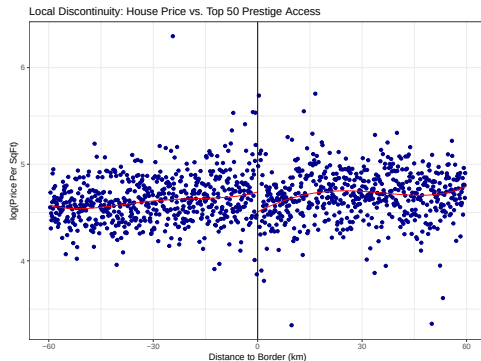
Next Research Steps

- **Show the ranking variation:** plot how ranks and Top 50 status change over time across treated border dyads.
- **Measure prestige more richly:** compare Top 50 counts with average rank, enrollment-weighted rank, and admissions-access measures.
- **Test mechanism heterogeneity:** estimate larger effects in counties with younger populations, more families, or higher expected college demand.
- **Stress-test identification:** add spillover checks, donut regressions, and a stronger institutional discussion of U.S. News ranking shocks.

Conclusion & Policy Implications

- **Summary Results:** Spatial equilibrium across US borders heavily prices higher-education prestige, completely neutralizing standard in-state tuition financial savings.
- **Implication for Policymakers:**
 - ▶ Broadly subsidizing generic higher education does *not* attract cross-border migration or inflate the local tax base.
 - ▶ True geographic brain-drain prevention and profound housing capitalization require sustained, concentrated investment into an elite, globally-recognized flagship capable of breaching the Top 50 threshold.

Visual Identification: The Prestige Discontinuity



- **Identification:** Binned scatter plots comparing cross-border housing prices in segments with top-tier access gaps.
- **The Result:** A clear, structural "jump" in the price surface at the Top 50 prestige boundary ($d_{ic} = 0$).

Appendix A: Pure Price Elasticity Restriction

- Restricted the panel strictly to border segments where $\Delta \text{Top } 50 = 0$.
- Does the financial price surface when Caliber prestige is identical?

Table: Pure Price Elasticity: Constant Top 50 Caliber (Caliber Gap = 0)

	(1) Single Cluster	(2) Two-Way Cluster
Tuition Gap LATE (\$1k)	-0.002 (0.005)	-0.002 (0.016)
Effective Property Tax Rate	-19.989*** (2.484)	-19.989*** (3.652)
Log Real GDP Per Capita	0.032*** (0.004)	0.032*** (0.010)
Local Unemployment Rate	-0.035*** (0.006)	-0.035*** (0.007)
TAXSIM Covariates	✓	✓
Fixed Effects	Seg $\times$ Year	Seg $\times$ Year
Observations	72,999	72,999
R^2	0.831	0.831

Result: A precise, perfectly-identified zero effect for raw financial cost.

Appendix B: Marginal Tax Cliff Restriction

- Restricted panel strictly to dyads located in the lowest income tax differential quartile. Eliminates Tiebout income-tax haven sorting as a confounder.

Table: Marginal Tax Cliff Capitalization: Lowest Quartile Tax Differential

	(1) Single Cluster	(2) Two-Way Cluster
Caliber Gap LATE (Top 50)	0.088* (0.053)	0.088 (0.091)
Tuition Gap LATE (\$1k)	0.012 (0.009)	0.012 (0.028)
Effective Property Tax Rate	-24.835*** (1.882)	-24.835*** (4.544)
TAXSIM / Macro Covariates	✓	✓
Fixed Effects	Seg × Year	Seg × Year
Observations	33,325	33,325
R^2	0.827	0.827

Appendix C: Stability Across Timeframes

- Does the Remote Work / COVID housing boom rupture our parameters?

Table: Timeframe Capitalization Heterogeneity: Pre-COVID vs. COVID Era

	(1) Pre-COVID	(2) Post-COVID
Caliber Gap LATE (Top 50)	0.105*** (0.038)	0.067* (0.037)
Tuition Gap LATE (\$1k)	-0.015 (0.017)	-0.002 (0.011)
Effective Property Tax Rate	-19.727*** (3.736)	-17.915*** (2.993)
TAXSIM / Macro Covariates	✓	✓
Fixed Effects	Seg × Year	Seg × Year
Standard Errors	Two-Way	Two-Way
Observations	71,531	46,648
R^2	0.774	0.837

Result: Model handles secular shocks well; the Top 50 premium operates strongly across time, though it compresses slightly during national housing spikes.